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Diseño de un controlador para un sistema de segundo orden

Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R I_1(s) + L [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R \right) I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{Cs} \right) I_2(s)$$

$$V_e(s) = (Ls + 2R) \left(\frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R \right) I_2(s) \right) - (Ls + R) I_2(s)$$

$$V_e(s) = \left(\frac{(Ls + 2R)(Ls + \frac{1}{Cs} + 3R)}{Ls + R} - (Ls + R) \right) I_2(s)$$

$$(Ls + 2R) \left(Ls + 3R + \frac{1}{Cs} \right) = L^2 s^2 + s(2LR + 6R^2) + \frac{L}{C} + \frac{2R}{Cs}$$

$$(Ls + R)^2 = L^2 s^2 + 2LRs + R^2$$

$$V_e(s) = \frac{3CLR^2 + (s(2LR^2 + L) + 2R)}{L^2 + CLs} I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(R + \frac{1}{Cs}) I_2(s)}{I_2(s)} = \frac{CLR^2 + (CR^2 + L)s + R}{3CLR^2 + (s(2LR^2 + L) + 2R) I_2(s)}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} \frac{1}{s} \frac{(1 - CLR^3 + (CR^2 + L)s + R)}{3CLR^3 + (5CR^2 + L)s + 2R} = 1 - \frac{R}{2R} = \frac{1}{2} = 0.5V$$

Estabilidad en la D abierta y polos del sistema

$$R = 5 \times 10^3$$

$$a = 3CLR$$

$$L = 10 \times 10^{-6}$$

$$b = 5CR^2 + L$$

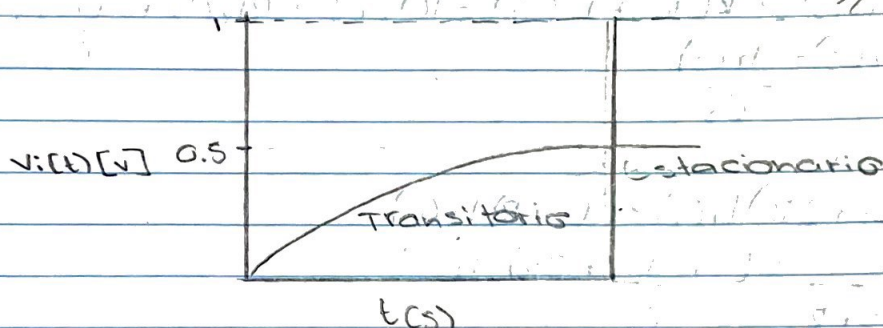
$$C = 470 \times 10^{-12}$$

$$c = 2R$$

$$\lambda_1 = -0.176$$

$$\lambda_2 = -8.33 \times 10^8$$

Sistema estable con respuesta sobreamortiguada



Modelo de ecuaciones integro-diferenciales

El modelo se expresa despejando las variables dependientes, es decir, las corrientes:

$$V_e(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = \left[\frac{V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}}{2R} \right] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + R \cdot i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[\frac{L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt}{3R} \right] \frac{1}{3R}$$

$$i_1(t) = \left[\frac{V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}}{2R} \right] \frac{1}{2R}$$

$$i_2(t) = \left[\frac{L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt}{3R} \right] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$